

Session Border Controller (SBC) Troubleshooting Runbook

NetOps Engineering — Voice Infrastructure Team

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1. Overview

This runbook covers diagnostic and resolution procedures for common Session Border Controller failures across all sites (London, Frankfurt, New York, Manchester). The SBC is the critical demarcation point between the internal UC network and external PSTN/SIP trunks. Failures on the SBC directly impact external call routing for all trading desks.

2. SIP Registration Failures

2.1 Symptoms

- IP phones showing 'Unregistered' status in CUCM device inventory
- SBC logs showing 'Registration Timeout' or '401 Unauthorized' responses
- Users receiving 'No Service' on physical handsets
- CUCM Real-Time Monitoring Tool (RTMT) showing device registration drops

2.2 Common Causes

Cause	Frequency	Check
Expired TLS certificate on SBC edge	High	Check cert expiry: show crypto pki cert
CUCM SIP trunk misconfiguration	Medium	Verify SIP trunk security profile
Network ACL blocking SIP port 5061	Medium	Test: telnet <sbcb-ip> 5061
DNS resolution failure for SBC FQDN	Low	nslookup sbcb.internal from CUCM node
SBC registration throttling active	Low	Check: show sip-ua register status

2.3 TLS Certificate Renewal Procedure

When registration failures are caused by an expired TLS certificate:

Step 1: Confirm certificate expiry on the SBC CLI:

```
show crypto pki certificates verbose
```

Step 2: Generate a new CSR from the SBC:

```
crypto pki enroll <trustpoint-name>
```

Step 3: Submit CSR to the internal CA (ca.internal) and download the signed cert.

Step 4: Import the new certificate:

```
crypto pki import <trustpoint-name> certificate
```

Step 5: Verify devices re-register in CUCM RTMT within 5 minutes.

3. SIP 503 Service Unavailable

3.1 Symptoms

- Callers receiving fast busy or 'All circuits busy' message
- SBC logs showing outbound 503 responses
- SBC CPU utilisation above 85% (check: show processes cpu sorted)

3.2 SIP OPTIONS Flood Diagnosis

A common cause of 503 errors is an OPTIONS flood from a misconfigured downstream device. Standard SIP OPTIONS keepalive interval is 30 seconds. A misconfigured PBX may send OPTIONS every 100ms, generating 36,000 requests/hour and saturating the SBC CPU.

Identify the source on the SBC CLI:

```
debug ccsip messages | include OPTIONS
```

Apply rate limiting for the offending source IP:

```
ip access-list extended RATE-LIMIT-OPTIONS permit udp host 10.20.1.45 any eq 5060
```

4. Codec Negotiation Failures

4.1 Symptoms

- External calls dropping at exactly 30 seconds (RTP timeout)
- SBC logs showing 'No common codec found' during INVITE processing
- One-way audio on established calls

4.2 Supported Codecs and Priority Order

Codec	Bitrate	Use Case	Supported
G.711 ulaw	64 kbps	PSTN, North America	Yes
G.711 alaw	64 kbps	PSTN, Europe	Yes
G.729	8 kbps	WAN, bandwidth-limited	Yes
G.722	64 kbps	HD voice, internal only	Yes
G.723	6.3 kbps	Legacy only	No — disable

Always ensure G.711 (both ulaw and alaw) is included as a fallback in the SBC media profile. Removing G.711 from the allowed codec list will cause all external PSTN calls to fail codec negotiation, as most PSTN carriers offer G.711 only.

5. Escalation Matrix

Severity	Response Time	Escalate To	Contact
P1 — Critical	15 minutes	L3 Voice Engineer	voice-oncall@company.com
P2 — High	1 hour	L2 NetOps	netops@company.com
P3 — Medium	4 hours	L1 Support	servicedesk@company.com
P4 — Low	Next day	L1 Support	servicedesk@company.com